Interim Report: Symmetry-Based Augmentation for Quadrotor Hover Control

SS26 Drone Control Project

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Abstract

This report summarizes the symmetry-oriented reinforcement learning experiments completed so far for quadrotor hover control. We compare baseline PPO, augmentation-only variants, mirror-loss variants, and a new discretized yaw augmentation family (C_n). Results are reported from TensorBoard logs using the scalar `Train/mean_reward`, aggregated as the mean of the last 50 logged points per run. The strongest and most stable performance is currently obtained by augmentation-only methods, especially in the C_6–C_8 range.

1 Objective

The main objective is to evaluate whether symmetry-aware training improves data efficiency and final control quality relative to standard PPO in a hover task. The current hypothesis is that enforcing or exploiting yaw equivariance in data augmentation improves policy quality and robustness.

2 Experimental Setup

- Task family: quadrotor hover control (direct RL environment).
- Algorithm: PPO-based training pipeline with symmetry options.
- Primary logged metric: `Train/mean_reward`.
- Aggregation rule: for each run, compute

$$R_{\text{final}} = \frac{1}{K}\sum_{i=1}^{K} r_i, \quad K = \min(50, N_{\text{logged}}),$$

  where r_i are the final logged reward values.
- Reporting: mean and standard deviation of R_{final} across runs in each experiment group.

3 Compared Variants

- **baselines**: PPO without symmetry augmentation.
- **aug_only**: augmentation-only symmetry variant.

- **aug_only_c4**: fixed C_4 augmentation variant.
- **mirror_loss_only**: mirror loss without augmentation.
- **aug_plus_mirror_loss**: combined augmentation and mirror loss.
- **aug_mirror**: legacy/combined family (as recorded in logs).

4 Results: Family-Level Comparison

Table 1: Final reward summary by experiment family (mean of last 50 points per run).

Family	Runs	Mean	Std	Min	Max
baselines	11	127.11	4.60	117.06	132.31
aug_only	11	131.41	2.26	125.71	133.97
aug_only_c4	11	131.30	1.59	127.28	132.66
mirror_loss_only	3	124.44	4.59	118.61	129.82
aug_plus_mirror_loss	3	127.43	3.15	123.32	130.96
aug_mirror	12	122.74	13.42	98.85	133.97

Observation. Augmentation-only methods currently outperform baseline PPO by approximately 4.2 reward points on average, while also reducing run-to-run variance. Mirror-loss-centric settings in the current form do not yet match augmentation-only performance.

5 Results: Discretized Yaw Augmentation (C_n Ablation)

The new discretized-theta experiments are organized under `aug_only_theta` with classes C_4, C_6, C_8, and C_{12}.

Table 2: C_n ablation summary (3 runs per class).

Class	Runs	Mean	Std	Min	Max
C_4	3	130.85	1.05	129.36	131.65
C_6	3	131.29	0.41	130.73	131.70
C_8	3	131.28	1.25	130.02	132.98
C_{12}	3	126.90	2.84	123.36	130.33

Observation. The current best region appears around $n \in \{6, 8\}$, while C_{12} underperforms and shows higher variability, suggesting over-discretization may hurt optimization in the present setup.

6 Figures

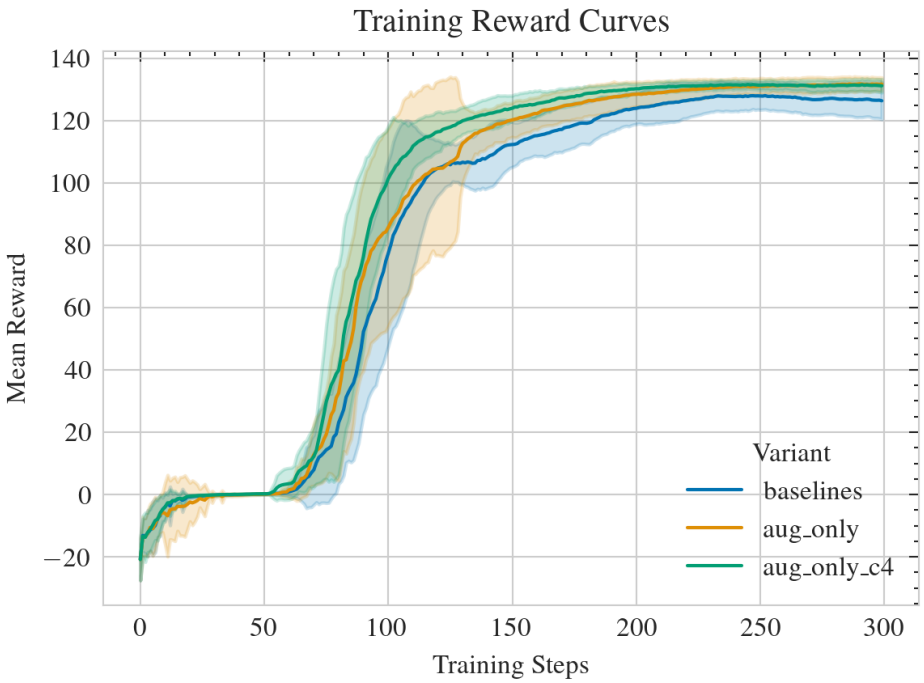


Figure 1: Training reward curves for baseline and symmetry families.

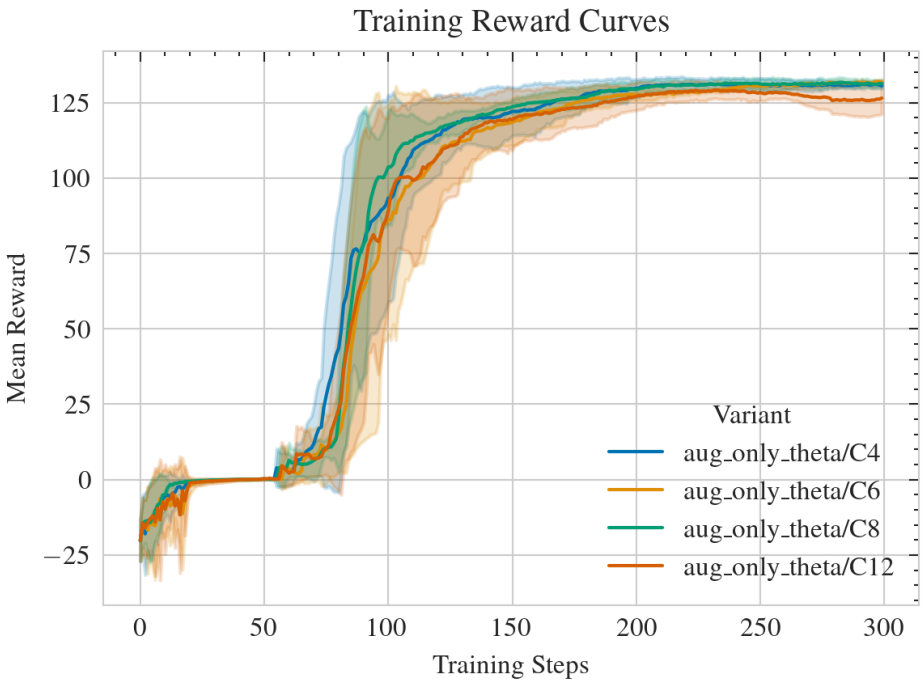


Figure 2: Training reward curves for discretized yaw classes (C_n).

7 Interpretation

- Symmetry augmentation is consistently beneficial over baseline in final training reward.
- Fixed-group augmentation (C_4) remains strong, but moderate discretization (C_6, C_8) may provide a small additional gain.
- Mirror-loss-heavy settings likely need further coefficient tuning and/or training-length tuning.

8 Limitations

- Some families use fewer runs (e.g., 3), making estimates less certain.
- The report uses a single scalar metric and does not yet include success-rate or robustness-to-disturbance evaluation.
- Hyperparameters are not fully controlled across all historical runs.

9 Planned Next Experiments

- Increase seeds for C_n variants to at least 10 per class.
- Perform a focused sweep around $n = 4, 6, 8, 10$ with fixed training budget.
- Revisit mirror loss with coefficient sweeps under the same training horizon.
- Add evaluation-time metrics (goal error distribution, disturbance recovery).